

3. DeepSeekMath-Style GRPO: KL Is in the Objective

This is the most important correction to make to the earlier GRPO explanation. In the DeepSeekMath formulation, the KL divergence penalty is represented **directly in the policy objective**. It should not be described as simply subtracting a KL value from the scalar reward before computing the group-relative advantage.

Rollout mechanics

For a prompt q , the rollout policy generates a group of G responses:

```
q -> {o_1, o_2, ..., o_G} -> verifier / reward -> {r_1, r_2, ..., r_G} -> group-relative advantages
```

The group rewards are normalized to produce the learning signal. A common conceptual form is:

$$A_i = (r_i - \text{mean}(r_1 \dots r_G)) / (\text{std}(r_1 \dots r_G) + \text{epsilon})$$

The exact implementation may add token-level details and normalization choices, but the central idea is unchanged: the group supplies a baseline instead of a separate learned value model.

Where the KL term enters

The DeepSeekMath-style objective combines a PPO-like clipped policy term with a KL penalty against a frozen reference policy. Conceptually:

```
Objective ~= mean_i[ min(ratio_i * A_i, clip(ratio_i, 1-eps, 1+eps) * A_i) ] - beta * D_KL(policy_theta || policy_ref)
```

Here, ratio_i is the policy-update ratio against the rollout/old policy, while the KL term compares the updated policy with the frozen reference policy. These are two different comparisons and should not be conflated.